

ECO 202 — In-Class Exam 1

Statistics and Data Analysis for Economics | Princeton University

September 23, 2026 | Classes 1–5 | 100 points

Name: _____

Student ID: _____

Timing. Aim to complete your answers in 55–60 minutes, leaving 10–15 minutes to check within the 70-minute target. Follow announced start/end times within the 80-minute meeting.

Conditions. No books, notes, calculators, AI, or other electronics; University-authorized accommodations apply. Fractions, radicals, and equivalent exact expressions are acceptable. Show reasoning in written answers. Data are teaching examples unless identified otherwise.

Points. Questions 1–4: one answer A–D, 5 points each, no negative marking. Questions 5–8: 20 points each; subpart points are shown. Label any continuation clearly.

Contents and suggested pacing

Question	Topic	Page	Points	Minutes
1	Observational units and variable types	2	5	2.5
2	Standardization and transformations	2	5	2.5
3	Correlation and a fitted line	2	5	2.5
4	Association and causation	2	5	2.5
5	Describing a distribution	3	20	12
6	Units and standardization	5	20	10
7	A descriptive regression	7	20	15
8	Treatment effects and counterfactuals	9	20	12
Checking and review			—	11
Total			100	70

Princeton University Honor Code

At the end of the examination, write the pledge in full below and sign your name.

“I pledge my honor that I have not violated the Honor Code during this examination.”

Signature: _____

Multiple choice (20 points; about 10 minutes)

1. A file has one row for each shop in each month and an industry code such as 101 or 202. Which statement is correct?
 - A. The unit is a shop; industry code is quantitative.
 - B. The unit is a shop-month; industry code is categorical.
 - C. The unit is a month; industry code is quantitative.
 - D. The unit is an industry; all numerical entries are quantitative.
2. An observation has standardized value $z = 3/2$. Every value in the dataset is transformed by $y = 10 - 2x$. What is the observation's standardized value in the transformed dataset?
 - A. $3/2$
 - B. $10 - 3$
 - C. $-3/2$
 - D. -3
3. A scatterplot has sample correlation zero, with both sample standard deviations positive. Which conclusion is justified?
 - A. There cannot be a relationship between the variables.
 - B. The fitted least-squares line with an intercept is horizontal.
 - C. The variables cannot be causally related.
 - D. Every observed response equals its sample mean.
4. An observational comparison finds higher earnings among workers who took a training course. Which additional fact alone establishes that training caused the difference?
 - A. None of the other three facts alone establishes causation.
 - B. Training happened before earnings were measured.
 - C. The earnings difference is large.
 - D. A second analyst reproduces the same difference.

5. Describing a distribution (20 points; about 12 minutes)

A quantitative variable is observed for six units and takes the values

2, 4, 5, 7, 8, 13.

Use the median-of-halves convention: split the sorted observations into equal lower and upper halves and take each half's median as its quartile.

(a) (10 points) Find the median, Q_1 , Q_3 , and IQR. Sketch a boxplot with a labeled numerical axis and whiskers extending to the most extreme observations within the 1.5IQR fences.

(b) (4 points) Calculate both fences. Is 13 flagged as an outlier?

- (c) **(6 points)** Replace only the largest value to obtain a dataset with mean 20 and the same median and IQR. Give the new largest value. Verify that the new mean is 20 and that the median and IQR are unchanged.

6. Units and standardization (20 points; about 10 minutes)

A fictional dataset records workers' hourly wages in dollars. The sample mean is $\bar{x} = 24$ and the sample standard deviation is $s_x = 4$.

- (a) **(4 points)** One worker earns 30 dollars per hour. Calculate this worker's z-score and interpret it in words.
- (b) **(4 points)** What hourly wage corresponds to a z-score of $-1/2$? Show your calculation and state the units.

- (c) **(6 points)** Express every hourly wage in cents rather than dollars. Find the new sample mean and sample standard deviation, with units. Explain the scaling rule you use.
- (d) **(6 points)** Now standardize the original dollar wages using their sample mean and sample standard deviation. State the mean and sample variance of the standardized values. If the original wage distribution is right-skewed, what happens to its shape? Explain briefly using the effects of shifting and rescaling.

7. A descriptive regression (20 points; about 15 minutes)

For five stores, x is display area in hundreds of square feet and y is weekly sales in thousands of dollars. The observed pairs are $(1, 2), (2, 4), (3, 5), (4, 4), (5, 5)$. The summaries are

$$\bar{x} = 3, \quad \bar{y} = 4, \quad S_{xx} = 10, \quad S_{yy} = 6, \quad S_{xy} = 6,$$

where $S_{xx} = \sum (x_i - \bar{x})^2$, $S_{yy} = \sum (y_i - \bar{y})^2$, and $S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y})$. You may use

$$b_1 = S_{xy}/S_{xx}, \quad b_0 = \bar{y} - b_1\bar{x}, \quad r = S_{xy}/\sqrt{S_{xx}S_{yy}}.$$

(a) (6 points) Find the fitted line with an intercept. Interpret its slope with units and appropriate descriptive wording.

(b) (4 points) Find the fitted value and residual for the store with $(x, y) = (4, 4)$. Interpret the residual's sign.

- (c) **(6 points)** Calculate r and r^2 . Interpret r^2 in terms of variation in weekly sales across these five stores.
- (d) **(4 points)** An analyst uses the line to predict sales at $x = 20$ and says increasing a store's area by one unit will cause sales to rise by b_1 . Identify the separate problems with these two uses of the fitted line.

8. Treatment effects and counterfactuals (20 points; about 12 minutes)

The following fictional table reveals both potential outcomes for each of four workers. $D_i = 1$ means the worker takes a specified training course; $Y_i(d)$ is subsequent earnings in thousands of dollars under treatment status d .

Worker	$Y_i(0)$	$Y_i(1)$	D_i
A	2	4	1
B	6	9	1
C	3	7	0
D	7	8	0

(a) (8 points) Give each worker's observed outcome and individual treatment effect.

(b) (8 points) Calculate the average treatment effect across all four workers and the observed treated-minus-untreated difference in mean earnings.

- (c) (4 points) For each worker, identify which potential outcome would be unobserved in real data. Could you calculate that worker's individual treatment effect from the observed outcome alone? Explain.